TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Songhak Yoon	Email	song.hak.yoon@iwks.fraunhofer.de
Institute Name	Fraunhofer IWKS			
Address	Aschaffenburger Str. 121, 63457 Hanau, Germany			

2. Technology Overview

Title of Proposed Technology	Development of high-purity lithium extraction technology through selective recovery of lithium contained in waste saggers			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Energy & Resources	
	Medium Categor	Small Category	Medium Category	Small Category
	Metal materials	Metal Refining/Recovery Technology	Resource Circulation	Inorganic Resource Recovery
Concept of Proposed Technology	■ Development of resource recovery technology enabling selective recovery of high-purity lithium compounds adsorbed and retained on the surface and pore layer of saggers discarded after repeated use in secondary battery cathode material manufacturing processes, alongside application of residual ceramics as refractories, additives, and other recyclable resources for other industries - Based on an estimated 15 uses per sagger for sintering, approximately 10 or more saggers are consumed and discarded per ton of secondary battery cathode material produced. The valuable metals within these waste saggers contain valuable resources: 95% ceramic, 1-2% lithium, and 2-3% other valuable metals. - Lithium compounds penetrating the surface and pore layer of waste saggers become physically fixed internally during high-temperature sintering. Due to chemical bonding with ceramic components like Al_2O_3 and SiO_2, they cannot be removed by simple washing or mechanical crushing - When the entire sagger is processed into raw material through simple crushing, the lithium content is only about 1-2%, making it difficult to secure production volume relative to the input raw material. Ensuring process			

efficiency and productivity is essential.

- Securing distribution data of residual lithium within spent batteries, implementing precision surface polishing to recover only lithium from high-purity zones, and commercializing technology to selectively recover lithium-concentrated lithium-ceramic composite powder through grinding equipment is essential.

[신품 Sagger]

[사용 후 폐기된 Sagger]

■ Development of Technology for Stabilizing Properties and Utilizing Recycled Resources in Residual Al_2O_3/SiO_2 Ceramics After Lithium Removal

- Waste sagger contains ceramic components averaging over 70% Al_2O_3; when physical and chemical stability is secured, it can be utilized as an industrial material requiring heat resistance, impact resistance, and chemical resistance
- After removing residual lithium, it can be re-resourced as a refractory additive, metal smelting flux additive, or cement raw material through particle size control and sorting processes

■ Development of high-purity lithium extraction technology from selectively recovered lithium-ceramic mixed powder (5.0~7.0 wt% Li_2O)

- Lithium in waste battery black mass exists as $LiMO_2$ (NCM, NCA) or Li_2CO_3 residues, making it reactable with acids/bases and extractable via wet methods.

Rationale for Joint R&D Project

■ Why Joint R&D Projects Are Necessary

1. Limitations of Multidisciplinary, Multi-Process Convergence Technologies Rather Than Single Technologies

This R&D project is not merely a simple lithium recovery technology, but a highly complex convergence technology organically integrating the following composite processes. It is difficult for any single country or research institution to independently secure the entire cycle, making collaborative R&D with role-sharing among institutions possessing core technologies in each field essential.

① Analysis of lithium distribution within the microstructure and pores of spent saggers
② Precision physical separation (surface polishing/grinding) technology
③ Optimization of solid-state reaction-based sintering (calcination/reduction) conditions
④ Wet extraction and conversion to high-purity lithium compounds
⑤ Stabilization of residual ceramic properties and application as industrial materials

2. Technical Gap Due to the Unique Ore-Like Structure of Lithium in Spent Sagger

Unlike black mass from spent batteries, lithium within spent sagger exists in a mineral-like structure ($LiAlSi_2O_6$ series) bonded to $Al_2O_3 \cdot SiO_2$ ceramics. This represents an area inaccessible to existing domestic lithium recycling technology (wet processes centered on black mass). high-temperature solid-state reaction mechanism

clarification, and reaction selectivity control pose significant technical risks without joint research with overseas institutions possessing accumulated experience in lithium ore processing.

3. Gap between Laboratory Scale and Industrial Scale in High-Temperature Calcination/Conversion Processes

Selective activation and conversion of lithium from lithium-ceramic mixed powders requires precise control of calcination and reduction conditions (temperature, residence time, atmosphere, additives). However, this process exhibits significant behavioral differences between laboratory and industrial scales, making pilot experience and process simulation capabilities essential.

This means that technological maturity can only be secured through a joint verification system with overseas research institutions possessing long-term practical experience in high-temperature ceramic and mineral processes, rather than through domestic research alone.

4. Challenges Requiring Simultaneous Technology Demonstration and Compliance with International Regulations and Standards

This project requires consideration of the following regulations and standards beyond merely producing research outcomes. Addressing these requirements solely based on domestic standards is limited; international regulatory-friendly technology design from the initial stages is necessary through joint R&D with European research institutions.

- Compliance with EU battery regulations
- Meeting circular economy-based standards for byproduct re-resource utilization
- Verification of eco-friendly processes and demonstration of carbon reduction effects

5. The Necessity of Complementary Research Infrastructure Between Domestic and Overseas

The strengths of both sides are incomplete when pursued independently; only through joint R&D can the entire technology lifecycle be completed.

(Korea)

- Actual waste Sagger generation processes and industrial data
- Demonstration environment linked to secondary battery cathode material manufacturing
- Commercialization and business infrastructure

(Germany)

- Core technologies for high-temperature ceramics, refractories, and mineral processing
- Know-how in conversion and extraction processes based on lithium ores
- Advanced materials analysis capabilities (XRD, SEM-EDS, high-temperature reaction analysis)

6. The Need for Strategic Joint Development to Secure Technology Leadership and Enable Global Expansion

The technology for lithium recovery and ceramic resource recycling based on spent Sagger is an unexplored field where global standards have yet to be established. By jointly setting the direction for technology development, verification, and standardization from the initial stages through collaborative R&D, it is possible to expand this technology into the European and global battery materials industry and secure market leadership.

■ Background Requiring Cooperation with Germany

1. Germany's Competitiveness in Core Technologies for High-Temperature Ceramics, Refractories, and Mineral Resource Processing

Germany possesses world-leading technological capabilities in high-temperature ceramics, refractories, and mineral-based resource processing. It possesses world-leading technological capabilities, particularly in solid-state reaction-based sintering processes, lithium mineral conversion technology, ceramic microstructure analysis, and property control. Germany's accumulated core technologies and process know-how are essential for this project's core objectives: the selective separation of lithium from spent sagger and elucidating the extraction mechanism for ore-type lithium.

2. Direct technological linkage with lithium ore-based extraction technology

Since lithium within spent sagger has a crystalline structure similar to lithium ore rather than black mass, Germany's strengths in lithium ore calcination, leaching, and conversion process technologies can be applied and advanced. Particularly, deriving energy-efficient calcination conditions, controlling reaction selectivity, and scaling up processes are areas difficult to secure through domestic research alone, necessitating minimization of technical risks through international joint research.

3. Strategic Cooperation for Addressing EU Circular Economy and Battery Regulations

Germany leads EU battery regulations and circular economy policies. The re-resource utilization of secondary battery manufacturing by-products is directly linked to future European market entry and securing technological standards. Technologies developed through joint research with Germany enable EU regulation-friendly process design and certification compliance, and can be expanded into technology commercialization and joint demonstration with global battery material companies.

4. Complementary Technology Structure Between Korea and Germany

Korea's strengths lie in secondary battery cathode material manufacturing processes, on-site data from spent Sagger generation, demonstration infrastructure, and strong industrial linkages. Germany excels in process theory, high-temperature reaction analysis, and core ceramic/mineral processing technologies. Cooperation between the two countries represents an optimal partnership structure that goes beyond simple technology transfer, enabling the global standardization of novel resource circulation technologies based on secondary battery byproducts.

Key Words	1. Cathode Material 2. Sagger 3. Lithium 4. Recycling 5. Circular Resource

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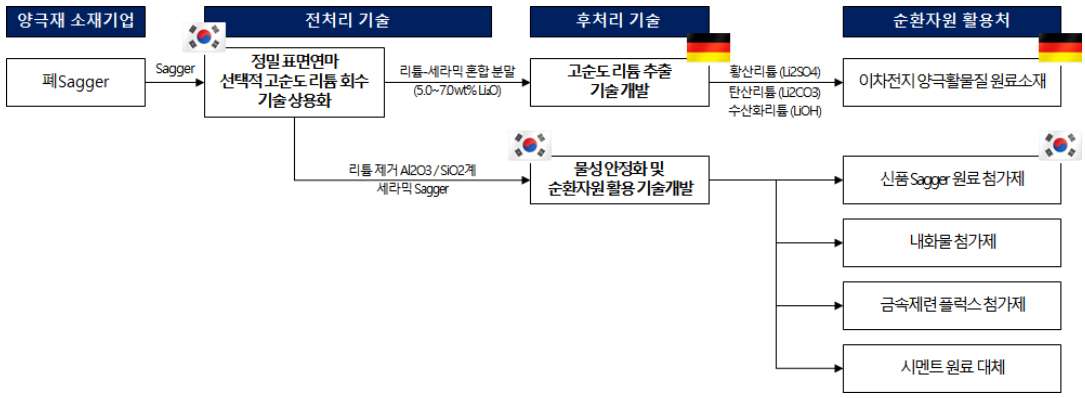
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	과제 시작 시 Start of the Project	Level (4)	과제 종료 시 End of the Project	Level (8)
R&D Objectives	■ This R&D project aims to develop optimal recycling technology and demonstrate commercialization for selectively recovering lithium resources accumulated on the surface of waste saggers generated after repeated use in the manufacturing process of secondary battery cathode materials, and re-resource them into high-value-added lithium compounds and recycled ceramic materials. 1. Acquisition of distribution data for residual lithium within waste saggers 2. Secure high-concentration lithium pre-treatment powder (≥ 4.5 wt%) through precision surface polishing and pilot-scale grinding equipment operation 3. Design and install a pilot-scale recycling facility for lithium-removed residual Al_2O_3/SiO_2-based saggers 4. Secure high-purity (≥99.0%) prototype samples of lithium sulfate, lithium carbonate, and lithium hydroxide 5. Secure prototype samples of Al_2O_3/SiO_2-based ceramic powder			
R&D Target Specifications	*Up on the discussion with Korean SMEs*			
R&D Plan	*Up on the discussion with Korean SMEs*			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	 [Conceptual Diagram for Development of Technology to Recover Lithium from Spent Sagger, Extract High-Purity Lithium, and Re-resource Residual Ceramic Materials]
Expected R&R of Korean SME Partner	Please refer to the diagram above.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Please refer to the diagram above.

6. References

(1)

(2)

□ Terms: 용어 설명

1) Industrial Technology Classification 산업기술 분류

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

대한민국 산업기술혁신사업 공통운영요령 제16조에 따른 산업기술분류표(별첨 1)에 근거한 분류체계

2) Technology Readiness Level (TRL) 기술성숙도 수준

Description 정의		
·A consistent and objective indicator of the technological maturity of key element technologies 핵심 요소 기술의 기술 성숙도를 일관되고 객관적으로 나타내는 지표		
	Basic Research Stage (1-2) **기초연구 단계 (1~2단계)**	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents 1단계: 기초 이론 및 실험 2단계: 개념 검증(PoC), 실용 아이디어 또는 특허 수준
	Experimental Stage (3-4) **실험 단계 (3~4단계)**	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale 3단계: 실험실 규모에서 기본 성능 검증 4단계: 실험실 규모에서 소재·부품·시스템의 핵심 성능 평가
	Prototype Stage (5-6) **시제품 단계 (5~6단계)**	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale 5단계: 최종화된 소재·부품·시스템의 시제품 제작 및 성능 평가 6단계: 파일럿 규모 시제품 제작 및 성능 평가
	Productization Stage (7-8) **제품화 단계 (7~8단계)**	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization 7단계: 잠재 고객 대상 신뢰성 시험 및 평가 8단계: 시제품 인증 및 표준화
	Commercialization Stage (9) **사업화 단계 (9단계)**	9th Stage: Commercialization 9단계: 상용화